List of Experiments2

Set 6:

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the

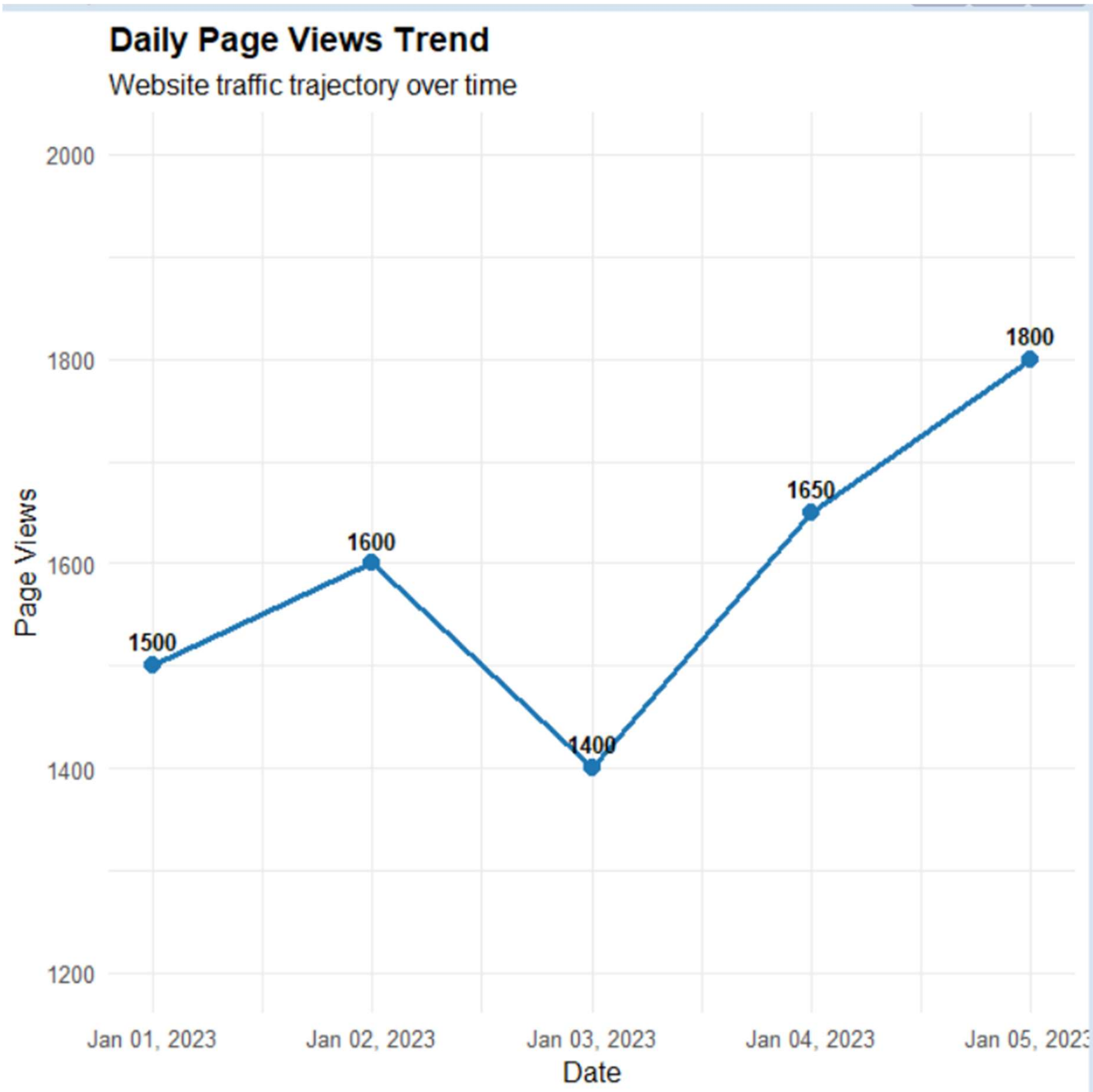
chart.

Code:

```
library(ggplot2)

library(dplyr)

library(tidyr)


# -----------------------------------------------------------

# Dataset Setup

# -----------------------------------------------------------

traffic_data <- data.frame(

  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-
05")),

  Page_Views = c(1500, 1600, 1400, 1650, 1800),

  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6),

  # Sample user interaction data for Task 3

  Likes = c(120, 145, 95, 130, 160),

  Shares = c(45, 55, 30, 50, 65),

  Comments = c(25, 32, 15, 22, 35)

)


# -----------------------------------------------------------

# Task 1: Line Chart (Daily Page Views Trend)

# -----------------------------------------------------------

ggplot(traffic_data, aes(x = Date, y = Page_Views)) +
```

```r
geom_line(color = "#1F77B4", size = 1.2) +
geom_point(color = "#1F77B4", size = 3) +
geom_text(aes(label = Page_Views), vjust = -0.8, fontface = "bold", size = 3.5) +
scale_x_date(date_labels = "%b %d, %Y", date_breaks = "1 day") +
scale_y_continuous(limits = c(1200, 2000)) +
labs(
  title = "Daily Page Views Trend",
  subtitle = "Website traffic trajectory over time",
  x = "Date",
  y = "Page Views"
) +
theme_minimal(base_size = 12) +
theme(plot.title = element_text(face = "bold", size = 14))
```

Output:



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart

elements.

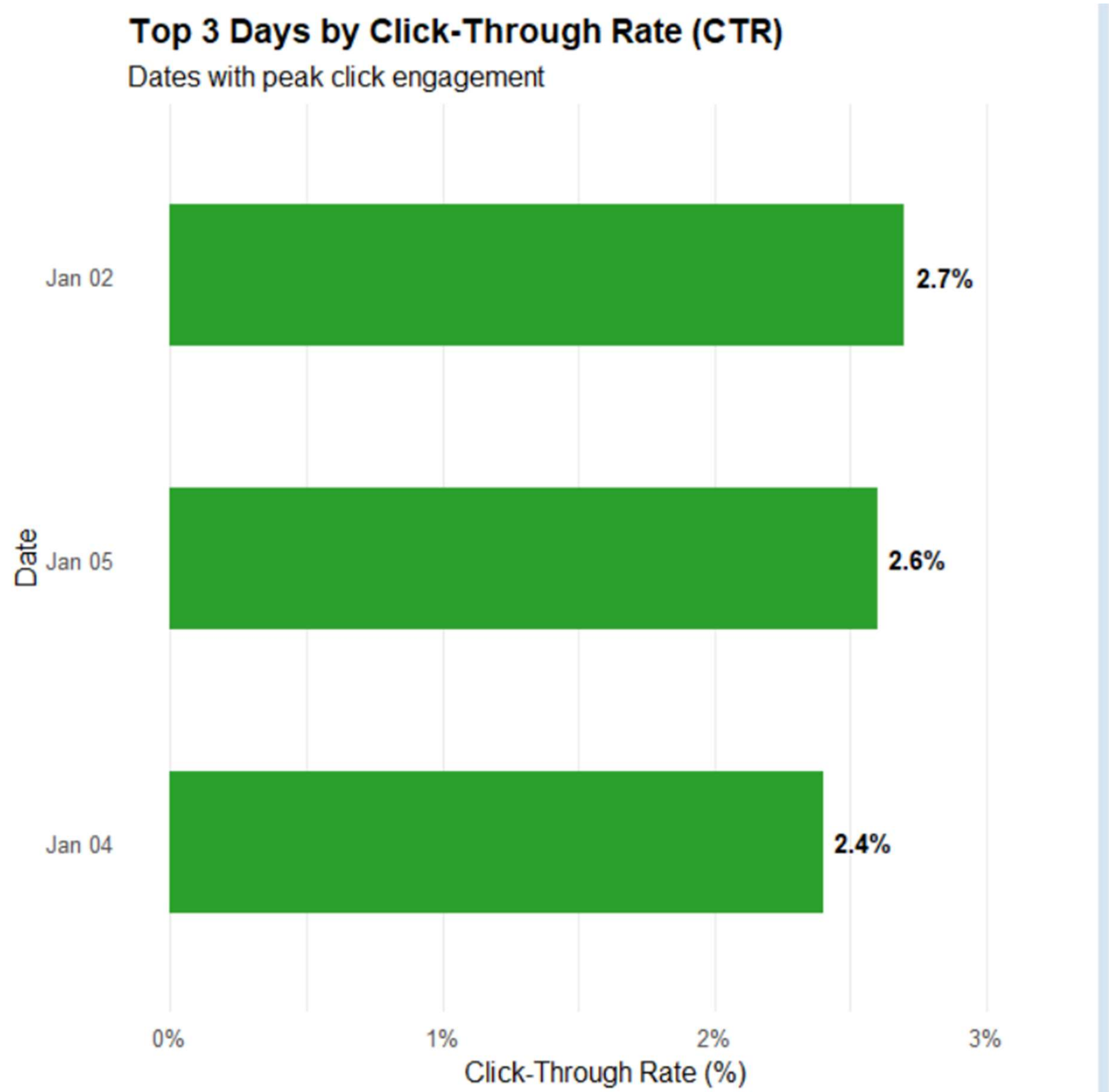
Code:

```
library(ggplot2)

library(dplyr)

library(tidyr)
```

```r
# ----------------------------------------------------------

# Dataset Setup

# ----------------------------------------------------------

traffic_data <- data.frame(

  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-
05")),

  Page_Views = c(1500, 1600, 1400, 1650, 1800),

  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6),

  # Sample user interaction data for Task 3

  Likes = c(120, 145, 95, 130, 160),

  Shares = c(45, 55, 30, 50, 65),

  Comments = c(25, 32, 15, 22, 35)

)


# ----------------------------------------------------------

# Task 2: Bar Chart (Top N Days by CTR)

# ----------------------------------------------------------

top_n_days <- 3  # You can adjust N here (e.g., 3 or 5)


traffic_data %>%

  slice_max(order_by = CTR, n = top_n_days) %>%

  ggplot(aes(x = reorder(format(Date, "%b %d"), CTR), y = CTR)) +

  geom_col(fill = "#2CA02C", width = 0.5) +

  geom_text(aes(label = paste0(CTR, "%")), hjust = -0.2, fontface = "bold", size = 3.8) +

  coord_flip() +

  scale_y_continuous(limits = c(0, 3.2), labels = function(x) paste0(x, "%")) +

  labs(

    title = paste("Top", top_n_days, "Days by Click-Through Rate (CTR)"),

    subtitle = "Dates with peak click engagement",
```

```
  x = "Date",
  y = "Click-Through Rate (%)"
) +
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(face = "bold", size = 14),
  panel.grid.major.y = element_blank()
)
```

Output:



3. Develop a stacked area chart to display the distribution of user interactions (likes, shares,

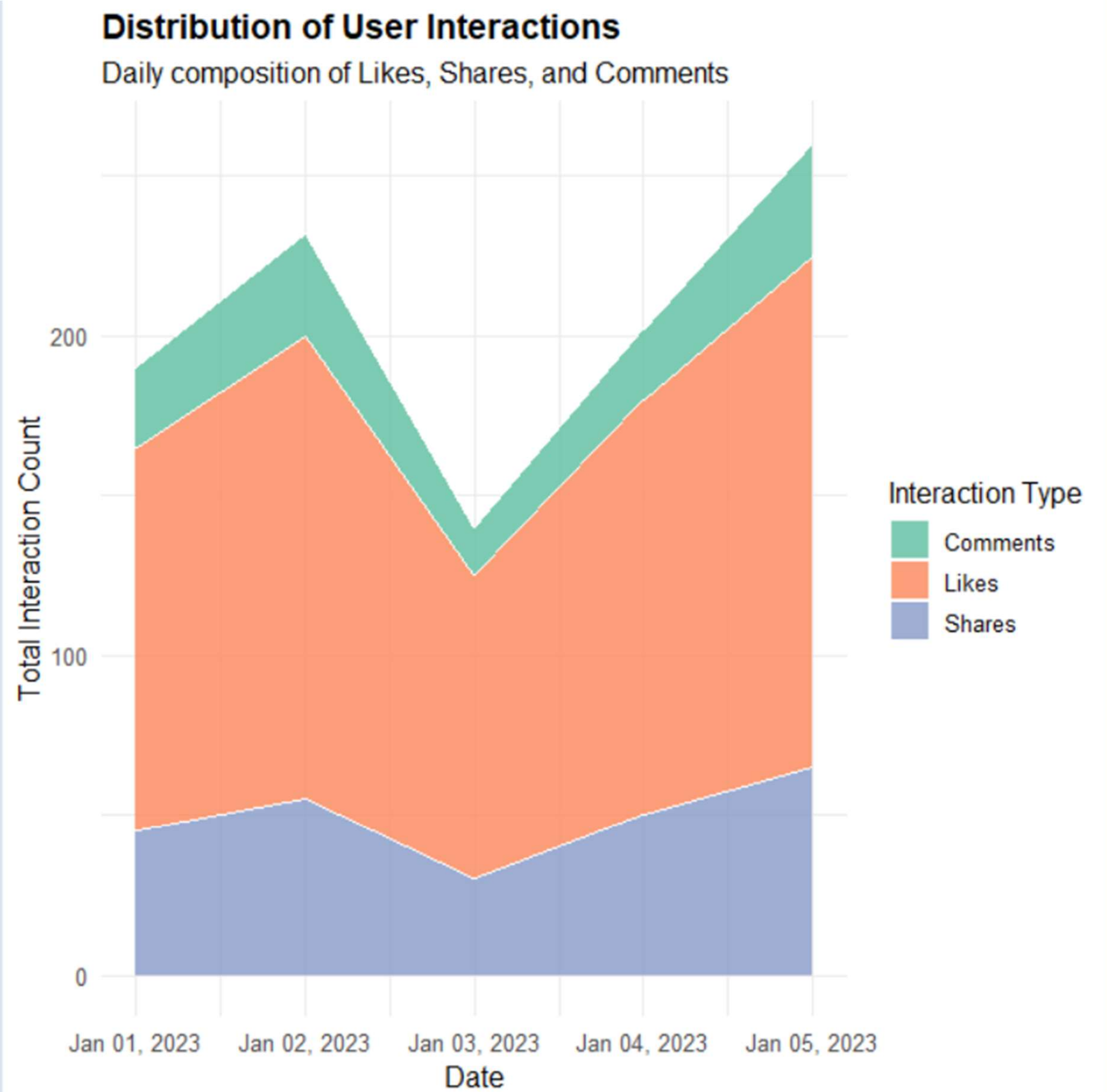
comments) on a website.

Code:

```
library(ggplot2)

library(dplyr)

library(tidyr)


# ------------------------------------------------------------

# Dataset Setup

# ------------------------------------------------------------

traffic_data <- data.frame(

  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-
05")),

  Page_Views = c(1500, 1600, 1400, 1650, 1800),

  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6),

  # Sample user interaction data for Task 3

  Likes = c(120, 145, 95, 130, 160),

  Shares = c(45, 55, 30, 50, 65),

  Comments = c(25, 32, 15, 22, 35)

)


# ------------------------------------------------------------

# Task 3: Stacked Area Chart (User Interactions Breakdown)

# ------------------------------------------------------------

# Convert wide data into long format for stacked plotting

interactions_long <- traffic_data %>%

  pivot_longer(

    cols = c(Likes, Shares, Comments),
```

```
    names_to = "Interaction_Type",
    values_to = "Count"
  )

ggplot(interactions_long, aes(x = Date, y = Count, fill = Interaction_Type)) +
  geom_area(alpha = 0.85, color = "white", size = 0.3) +
  scale_fill_brewer(palette = "Set2") +
  scale_x_date(date_labels = "%b %d, %Y", date_breaks = "1 day") +
  labs(
    title = "Distribution of User Interactions",
    subtitle = "Daily composition of Likes, Shares, and Comments",
    x = "Date",
    y = "Total Interaction Count",
    fill = "Interaction Type"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Output:



Set 7:

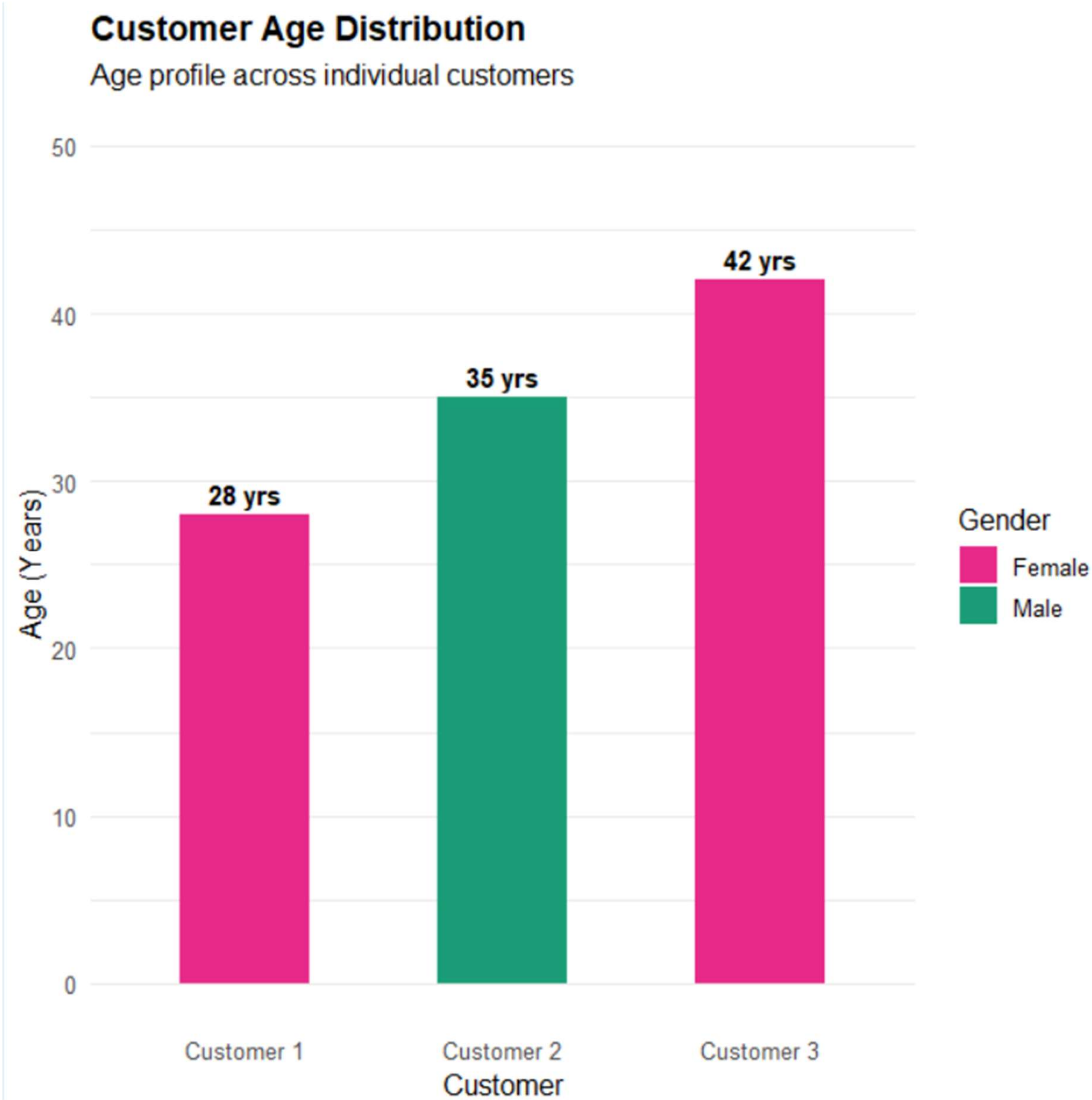
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Code:

```
# Load required libraries

library(ggplot2)

library(dplyr)


# Create the customer demographics dataset
```

```r
demographics_data <- data.frame(
  Customer_ID = factor(c(1, 2, 3)),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

# View the dataset
print(demographics_data)

# 1. Bar Chart: Customer Age Distribution
ggplot(demographics_data, aes(x = paste("Customer", Customer_ID), y = Age, fill =
Gender)) +
  geom_col(width = 0.5) +
  geom_text(aes(label = paste(Age, "yrs")), vjust = -0.5, fontface = "bold", size = 4) +
  scale_fill_manual(values = c("Female" = "#E7298A", "Male" = "#1B9E77")) +
  scale_y_continuous(limits = c(0, 50)) +
  labs(
    title = "Customer Age Distribution",
    subtitle = "Age profile across individual customers",
    x = "Customer",
    y = "Age (Years)",
    fill = "Gender"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.major.x = element_blank()
  )
```

Output:



2. Generate a pie chart to represent the distribution of customers by gender.

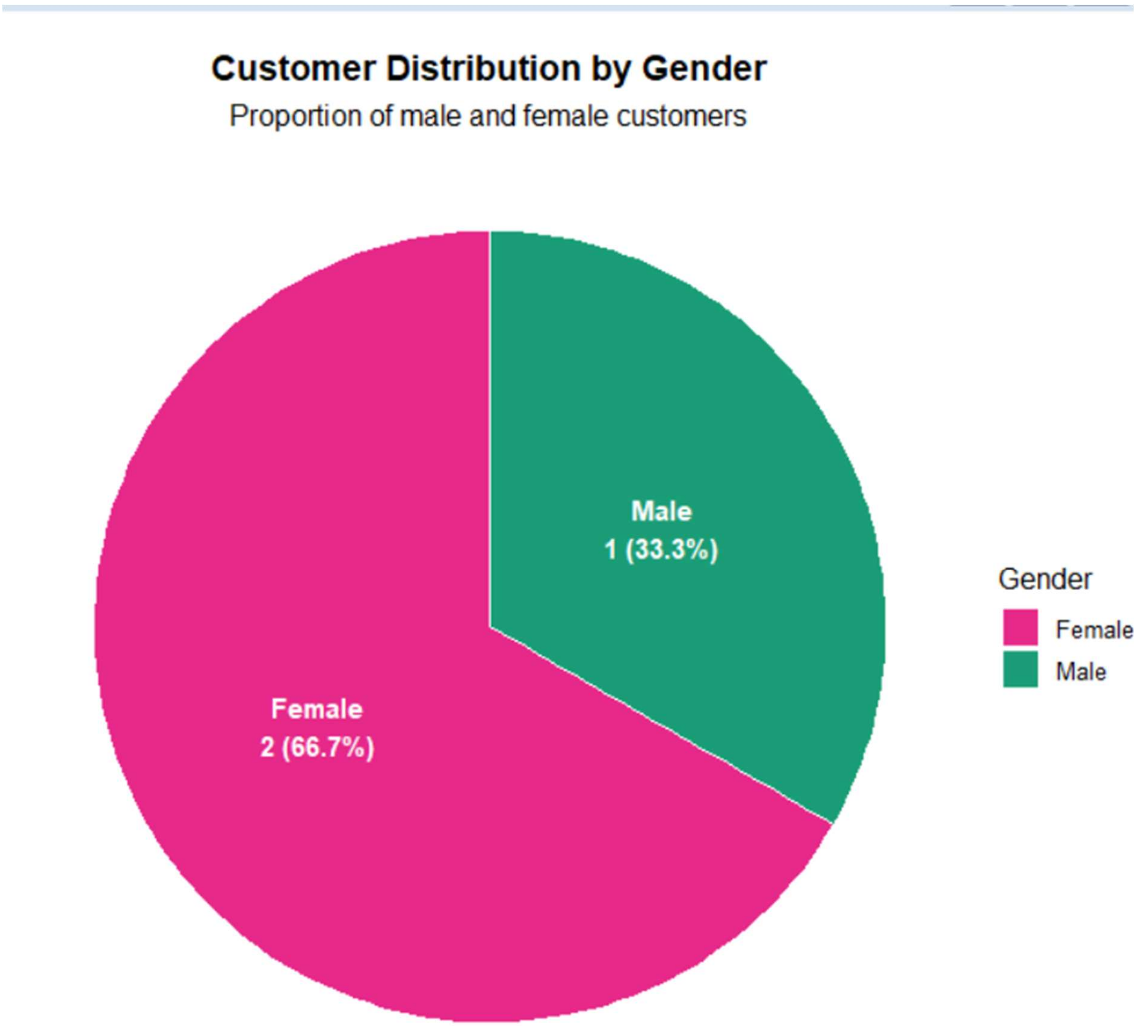
Code:

```
# Load required libraries

library(ggplot2)

library(dplyr)


# Create the customer demographics dataset

demographics_data <- data.frame(

  Customer_ID = factor(c(1, 2, 3)),
```

```r
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

# View the dataset
print(demographics_data)

# 2. Pie Chart: Gender Distribution
gender_counts <- demographics_data %>%
  count(Gender) %>%
  mutate(
    Percentage = n / sum(n) * 100,
    Label = paste0(Gender, "\n", n, " (", round(Percentage, 1), "%)")
  )

ggplot(gender_counts, aes(x = "", y = n, fill = Gender)) +
  geom_col(width = 1, color = "white") +
  coord_polar(theta = "y") +
  geom_text(aes(label = Label), position = position_stack(vjust = 0.5), color = "white",
fontface = "bold", size = 4) +
  scale_fill_manual(values = c("Female" = "#E7298A", "Male" = "#1B9E77")) +
  labs(
    title = "Customer Distribution by Gender",
    subtitle = "Proportion of male and female customers",
    fill = "Gender"
  ) +
  theme_void(base_size = 12) +
  theme(
```

```
  plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
  plot.subtitle = element_text(hjust = 0.5)
)
```

Output:



3. Build a table to show the demographic information for each customer. Label the table elements.

Code:

```
# Load required libraries
library(ggplot2)
```

```
library(dplyr)

# Create the customer demographics dataset
demographics_data <- data.frame(
  Customer_ID = factor(c(1, 2, 3)),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

# View the dataset
print(demographics_data)

# Load knitr for clean table formatting
library(knitr)

# Create a formatted table with column labels
demographic_table <- demographics_data %>%
  mutate(Income = paste0("$", format(Income, big.mark = ","))) %>%
  rename(
    `Customer ID` = Customer_ID,
    `Age (Years)` = Age,
    `Gender` = Gender,
    `Annual Income` = Income
  )

# Display formatted markdown table
kable(demographic_table, caption = "Customer Demographic Summary", align = "c")
```

Output:

```
Table: Customer Demographic Summary

| Customer ID | Age (Years) | Gender | Annual Income |
|:-----------:|:-----------:|:------:|:-------------:|
|      1      |      28     | Female |    $50,000    |
|      2      |      35     |  Male  |    $60,000    |
|      3      |      42     | Female |    $75,000    |
> |
```

Set 8:

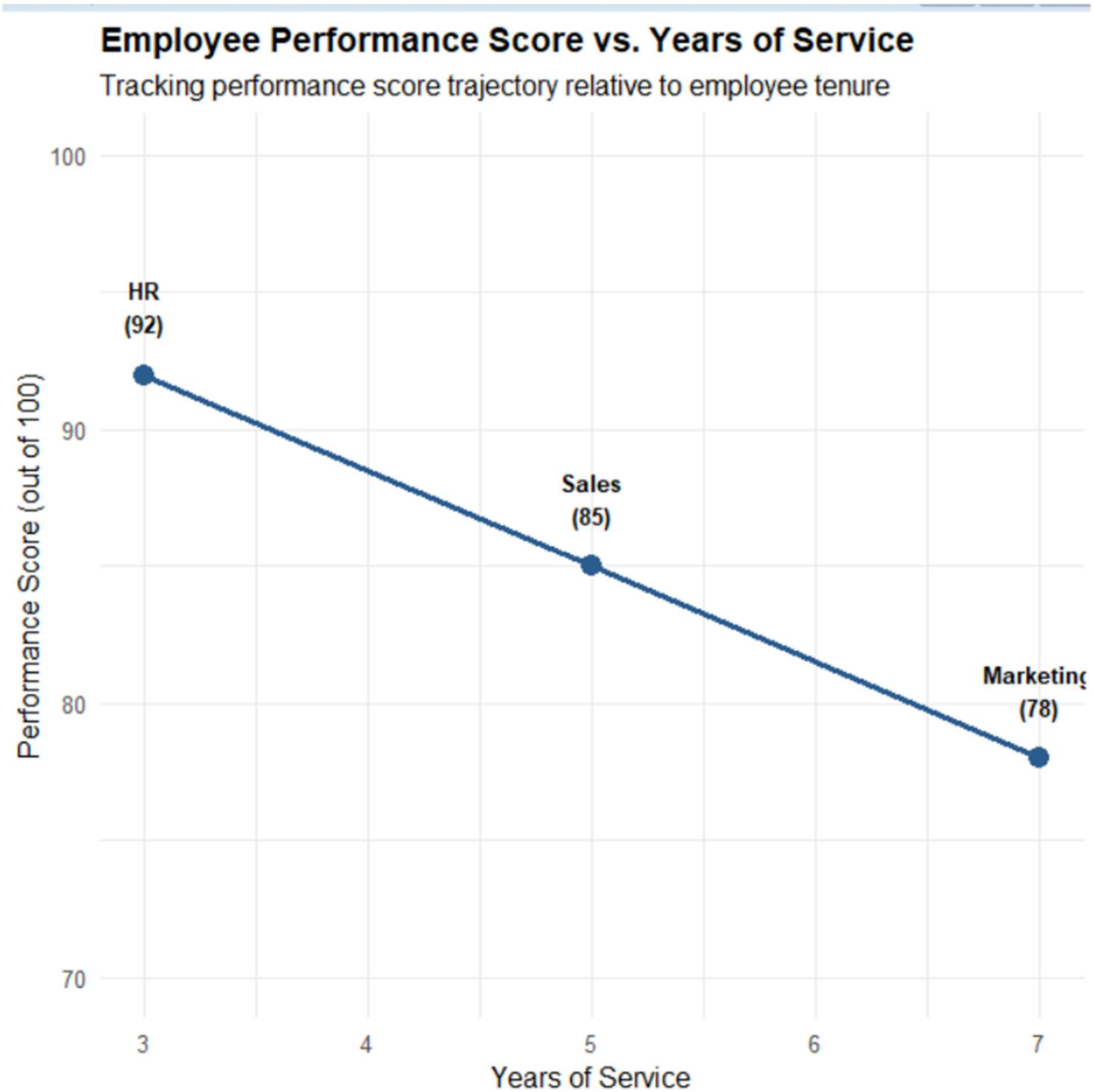
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and

title the chart.
Code:

```
# Load required libraries

library(ggplot2)

library(dplyr)


# Create the employee performance dataset

performance_data <- data.frame(

  Employee_ID = factor(c(1, 2, 3)),

  Department = c("Sales", "HR", "Marketing"),

  Years_of_Service = c(5, 3, 7),

  Performance_Score = c(85, 92, 78)

)


# Preview dataset

print(performance_data)


# 1. Line Chart: Performance Score over Years of Service

# Arrange by Years of Service for a chronological line progression

performance_data_sorted <- performance_data %>%

  arrange(Years_of_Service)
```

```
ggplot(performance_data_sorted, aes(x = Years_of_Service, y = Performance_Score)) +
  geom_line(color = "#2B5C8F", size = 1.2) +
  geom_point(color = "#2B5C8F", size = 4) +
  geom_text(
    aes(label = paste0(Department, "\n(", Performance_Score, ")")),
    vjust = -0.8,
    fontface = "bold",
    size = 3.5
  ) +
  scale_y_continuous(limits = c(70, 100)) +
  scale_x_continuous(breaks = seq(1, 8, by = 1)) +
  labs(
    title = "Employee Performance Score vs. Years of Service",
    subtitle = "Tracking performance score trajectory relative to employee tenure",
    x = "Years of Service",
    y = "Performance Score (out of 100)"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Output:



2. Generate a bar chart showing the distribution of employees across different departments. Label

the chart elements.

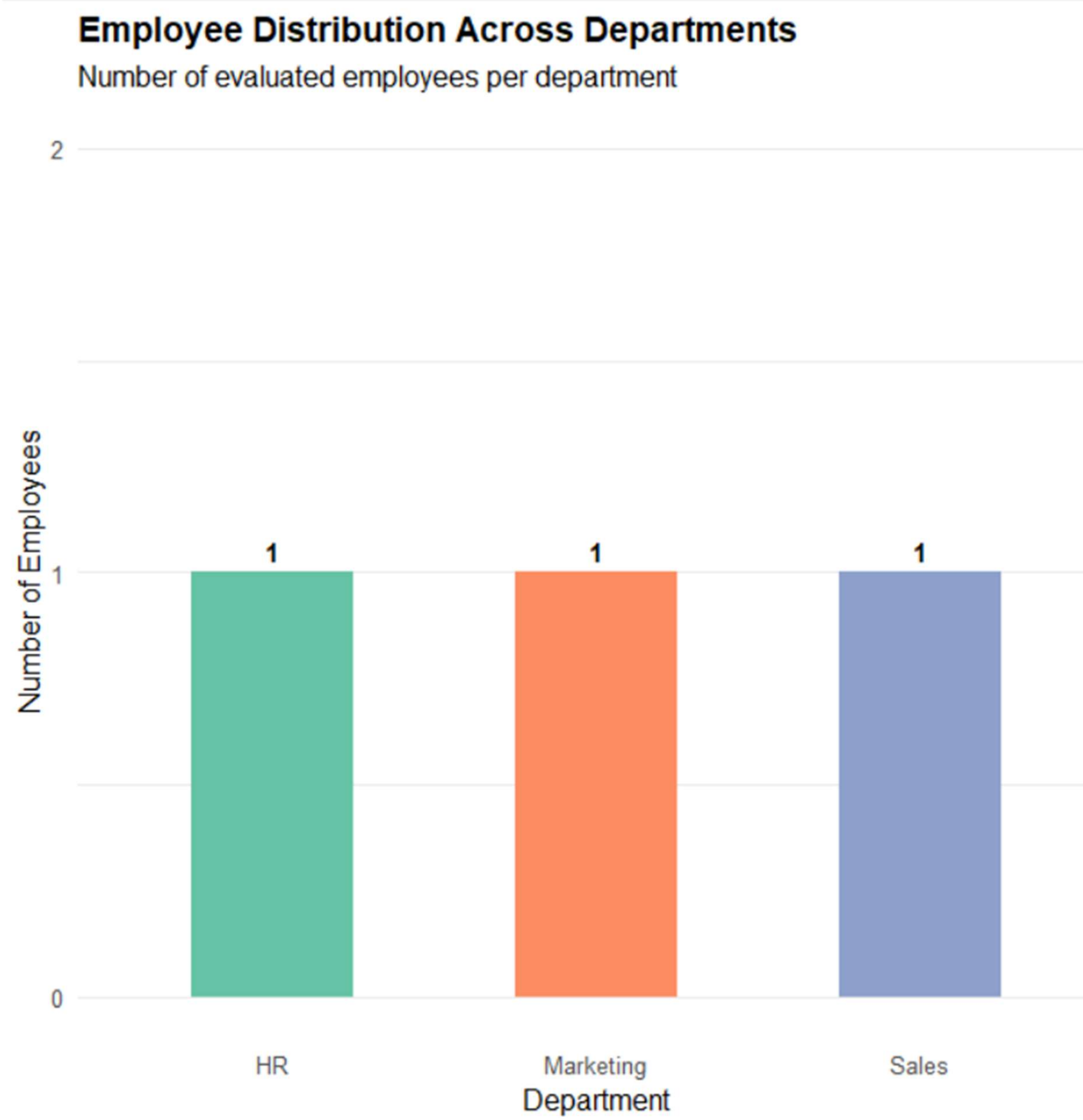
Code:

```
# Load required libraries
library(ggplot2)
library(dplyr)

# Create the employee performance dataset
performance_data <- data.frame(
```

```r
  Employee_ID = factor(c(1, 2, 3)),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Preview dataset
print(performance_data)

# 2. Bar Chart: Department Distribution
department_counts <- performance_data %>%
  count(Department)

ggplot(department_counts, aes(x = Department, y = n, fill = Department)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = n), vjust = -0.5, fontface = "bold", size = 4) +
  scale_y_continuous(limits = c(0, 2), breaks = c(0, 1, 2)) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Employee Distribution Across Departments",
    subtitle = "Number of evaluated employees per department",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.major.x = element_blank()
```

)

Output:



3. Build a table to display the performance data for each employee. Label the table elements.

Code:

```
# Load required libraries

library(ggplot2)

library(dplyr)


# Create the employee performance dataset
```

```r
performance_data <- data.frame(
  Employee_ID = factor(c(1, 2, 3)),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Preview dataset
print(performance_data)

# Load knitr for formatted table rendering
library(knitr)

# Format and label table columns
performance_table <- performance_data %>%
  rename(
    `Employee ID` = Employee_ID,
    `Department` = Department,
    `Years of Service` = Years_of_Service,
    `Performance Score` = Performance_Score
  )

# Output clean table
kable(performance_table, caption = "Employee Performance Summary Table", align = "c")
```

Output:

```
Table: Employee Performance Summary Table

| Employee ID | Department | Years of Service | Performance Score |
|:-----------:|:----------:|:----------------:|:-----------------:|
|      1      |   Sales    |        5         |        85         |
|      2      |     HR     |        3         |        92         |
|      3      | Marketing  |        7         |        78         |
> |
```

Set 8:

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and

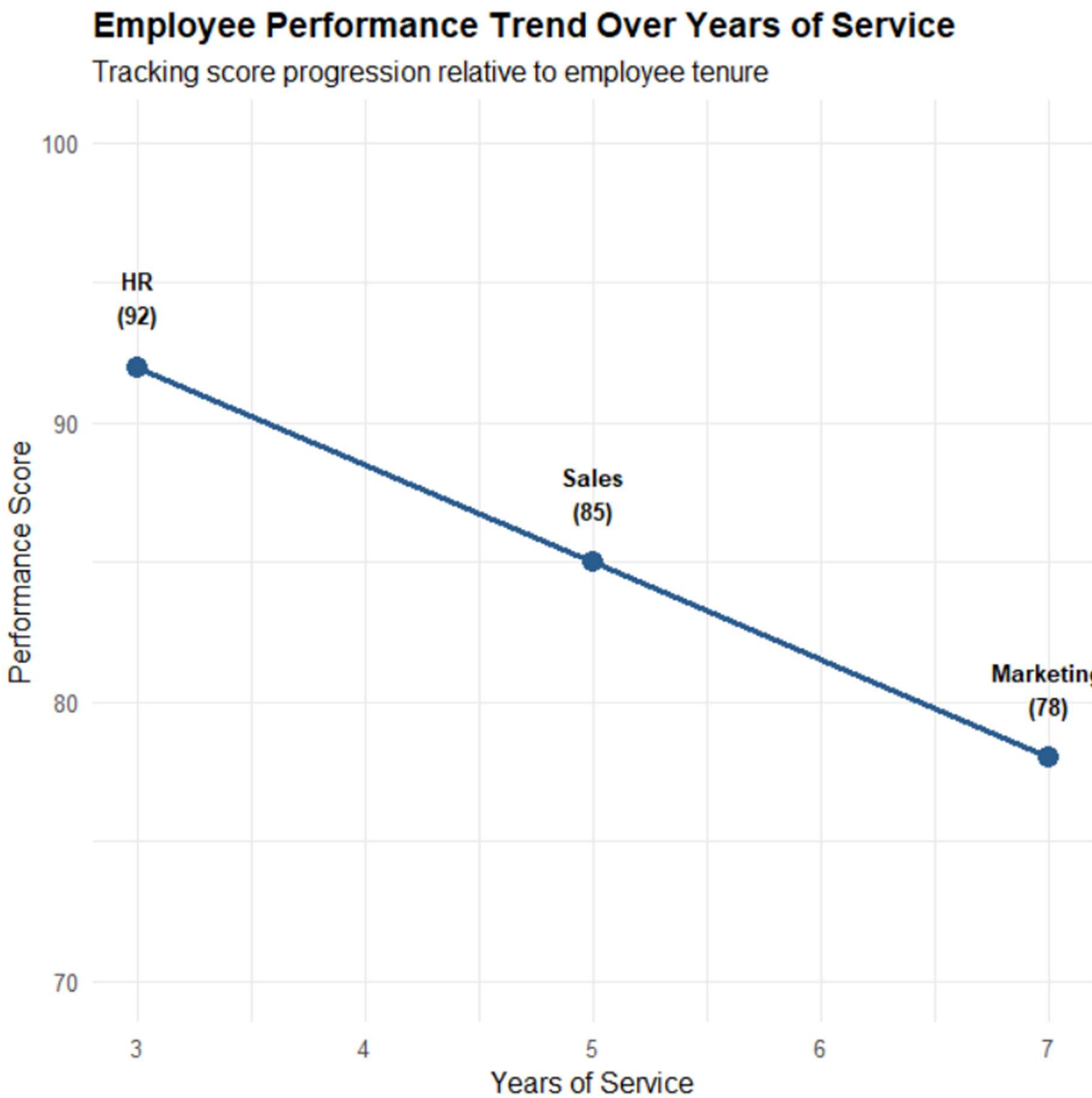
title the chart.

Code:

```
# ============================================================

# Set 8: Employee Performance Analysis — Complete Script

# ============================================================


# Load required libraries

library(ggplot2)

library(dplyr)

library(knitr)


# ------------------------------------------------------------

# Dataset Setup

# ------------------------------------------------------------

performance_data <- data.frame(

  Employee_ID = factor(c(1, 2, 3)),

  Department = c("Sales", "HR", "Marketing"),

  Years_of_Service = c(5, 3, 7),

  Performance_Score = c(85, 92, 78)

)
```

```r
# -----------------------------------------------------------
# Task 1: Line Chart (Performance Trend Over Time / Tenure)
# -----------------------------------------------------------
# Sort by tenure to plot chronological performance flow
performance_data_sorted <- performance_data %>%
  arrange(Years_of_Service)


ggplot(performance_data_sorted, aes(x = Years_of_Service, y = Performance_Score)) +
  geom_line(color = "#2B5C8F", size = 1.2) +
  geom_point(color = "#2B5C8F", size = 4) +
  geom_text(
    aes(label = paste0(Department, "\n(", Performance_Score, ")")),
    vjust = -0.8,
    fontface = "bold",
    size = 3.5
  ) +
  scale_y_continuous(limits = c(70, 100)) +
  scale_x_continuous(breaks = seq(1, 8, by = 1)) +
  labs(
    title = "Employee Performance Trend Over Years of Service",
    subtitle = "Tracking score progression relative to employee tenure",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Output:



2. Generate a bar chart showing the distribution of employees across different departments. Label

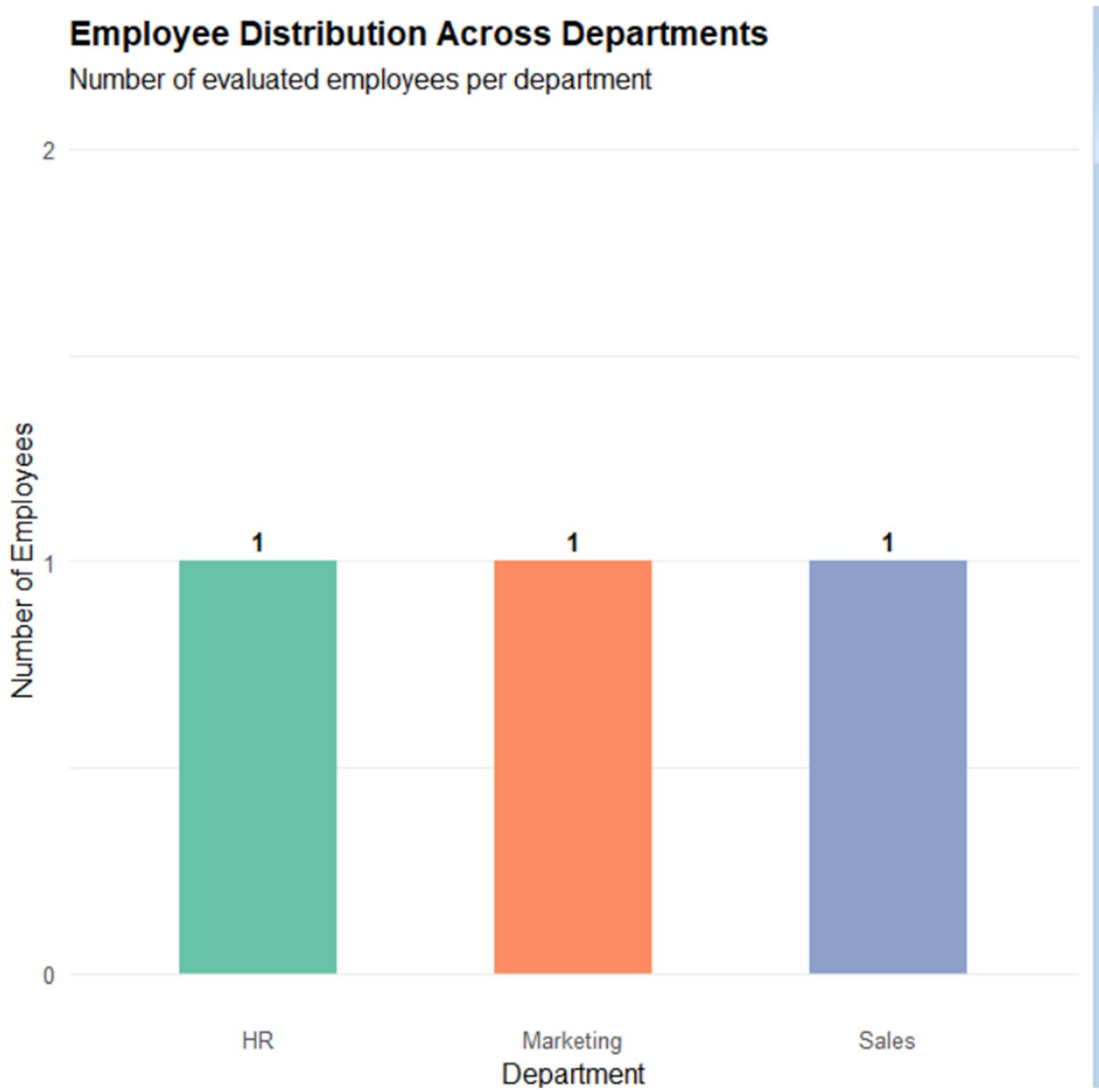
the chart elements.
Code:

```
# ============================================================

# Set 8: Employee Performance Analysis — Complete Script

# ============================================================


# Load required libraries
```

```
library(ggplot2)
library(dplyr)
library(knitr)

# -----------------------------------------------------------
# Dataset Setup
# -----------------------------------------------------------
performance_data <- data.frame(
  Employee_ID = factor(c(1, 2, 3)),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# -----------------------------------------------------------
# Task 2: Bar Chart (Employee Distribution Across Departments)
# -----------------------------------------------------------
department_counts <- performance_data %>%
  count(Department)

ggplot(department_counts, aes(x = Department, y = n, fill = Department)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = n), vjust = -0.5, fontface = "bold", size = 4) +
  scale_y_continuous(limits = c(0, 2), breaks = c(0, 1, 2)) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Employee Distribution Across Departments",
    subtitle = "Number of evaluated employees per department",
    x = "Department",
```

```
  y = "Number of Employees"
) +
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(face = "bold", size = 14),
  panel.grid.major.x = element_blank()
)
```

Output:



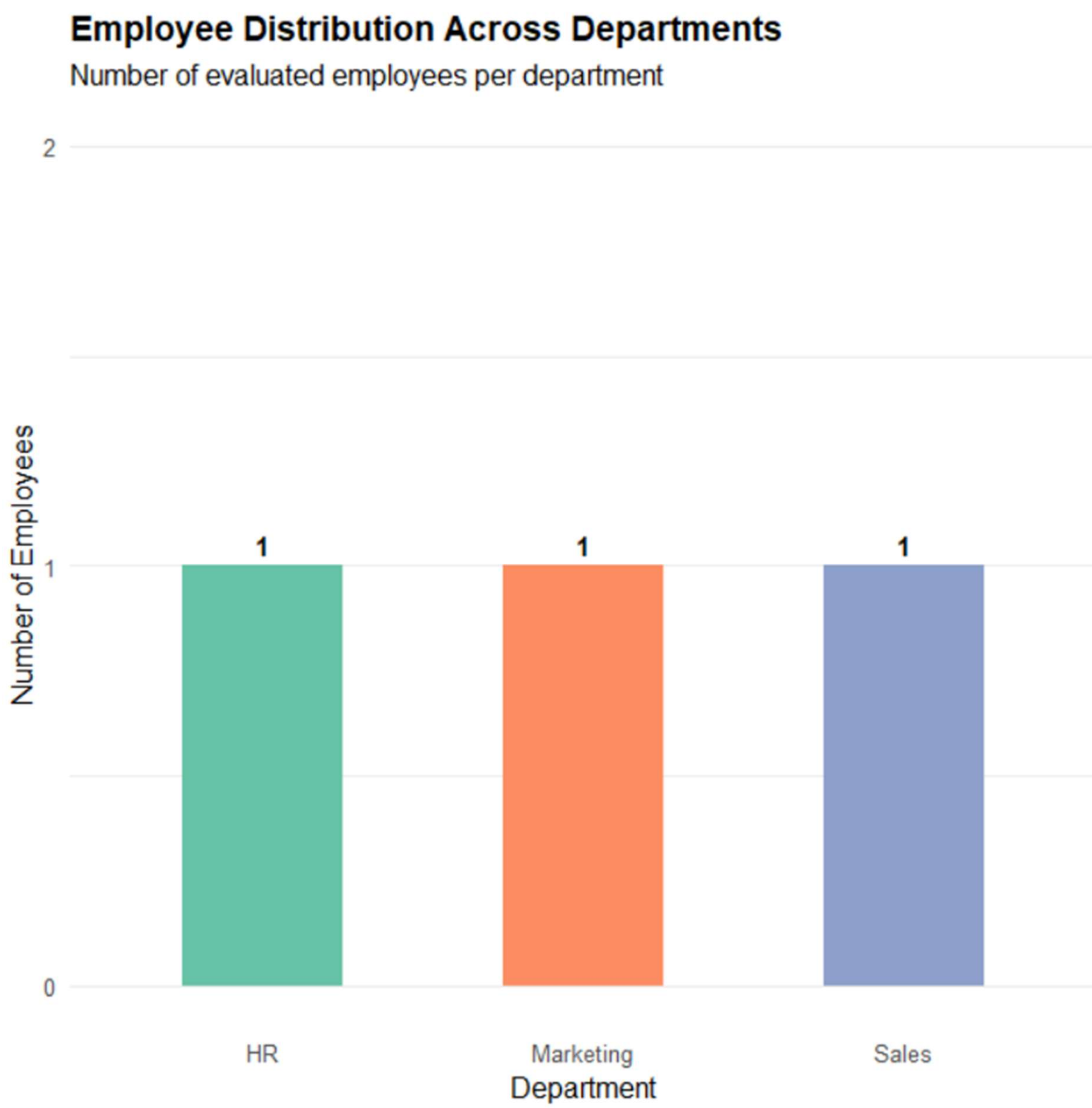
3. Build a table to display the performance data for each employee. Label the table

elements.
Code:

```
# ================================================================
# Set 8: Employee Performance Analysis — Complete Script
# ================================================================

# Load required libraries
library(ggplot2)
library(dplyr)
library(knitr)

# ------------------------------------------------------------
# Dataset Setup
# ------------------------------------------------------------
performance_data <- data.frame(
  Employee_ID = factor(c(1, 2, 3)),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# ------------------------------------------------------------
# ------------------------------------------------------------
# Task 3: Performance Data Table
# ------------------------------------------------------------
performance_table <- performance_data %>%
  rename(
    `Employee ID` = Employee_ID,
    `Department` = Department,
```

```
  `Years of Service` = Years_of_Service,
  `Performance Score` = Performance_Score
 )

# Print clean table format
kable(performance_table, caption = "Employee Performance Summary Table", align = "c")
```

Output:



Set 9:

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by

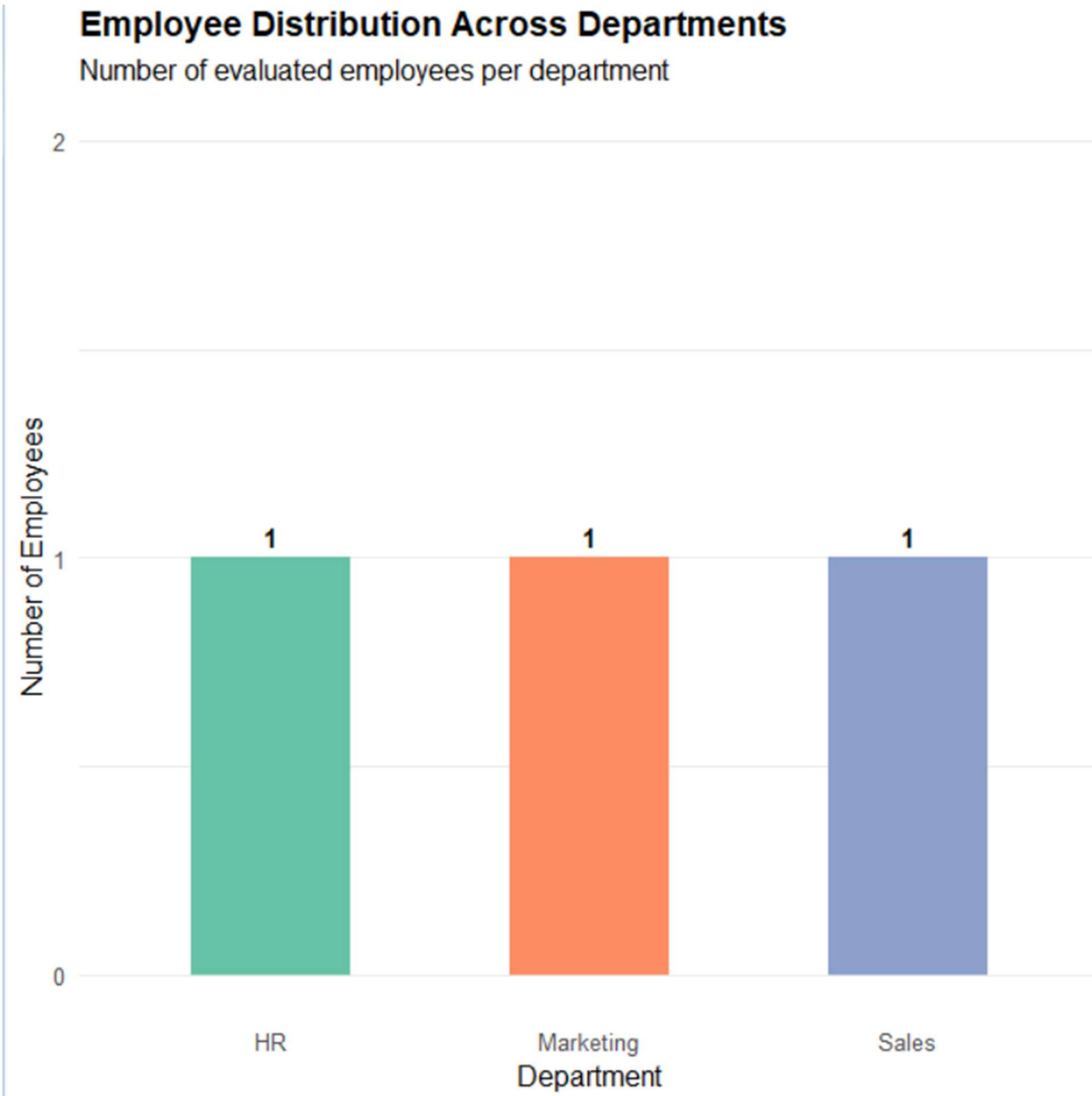
Course. Add title and labels. Interpret student performance.
Code:

```
# Load required libraries

library(ggplot2)

library(dplyr)

library(zoo) # For moving average calculations


# 1. Import dataset from CSV or build directly

# learning_data <- read.csv("Online_Learning_Activity.csv")


learning_data <- data.frame(

  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),

  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),

  Age = c(20, 22, 19, 21, 23, 20),

  Course = c("R", "R", "SQL", "R", "R", "SQL"),

  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),

  Videos_Watched = c(12, 15, 8, 18, 9, 14),

  Quiz_Score = c(78, 85, 65, 92, 70, 88),

  Login_Date = c("2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-03-12",
"2025-03-12")

)


# View dataset structure

str(learning_data)


# ----------------------------------------------------------

# 1a. Histogram of Quiz Scores

# ----------------------------------------------------------
```

```r
ggplot(learning_data, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 5, fill = "#2B5C8F", color = "white", alpha = 0.85) +
  geom_text(stat = "bin", binwidth = 5, aes(label = ifelse(..count.. > 0, ..count.., "")), vjust = -0.5, fontface = "bold") +
  scale_y_continuous(limits = c(0, 3), breaks = 0:3) +
  labs(
    title = "Distribution of Student Quiz Scores",
    subtitle = "Overall score distribution across all courses",
    x = "Quiz Score",
    y = "Frequency (Number of Students)"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14))


# ------------------------------------------------------------
# 1b. Boxplot of Quiz Score by Course
# ------------------------------------------------------------
ggplot(learning_data, aes(x = Course, y = Quiz_Score, fill = Course)) +
  geom_boxplot(alpha = 0.7, outlier.color = "red", width = 0.4) +
  geom_jitter(width = 0.05, size = 3, alpha = 0.8) +
  scale_fill_manual(values = c("R" = "#3182BD", "SQL" = "#E6550D")) +
  labs(
    title = "Quiz Score Performance by Course",
    subtitle = "Comparison of score distribution between R and SQL courses",
    x = "Course Name",
    y = "Quiz Score",
    fill = "Course"
  ) +
  theme_minimal(base_size = 12) +
```

```
theme(plot.title = element_text(face = "bold", size = 14))
```

Output:



2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent

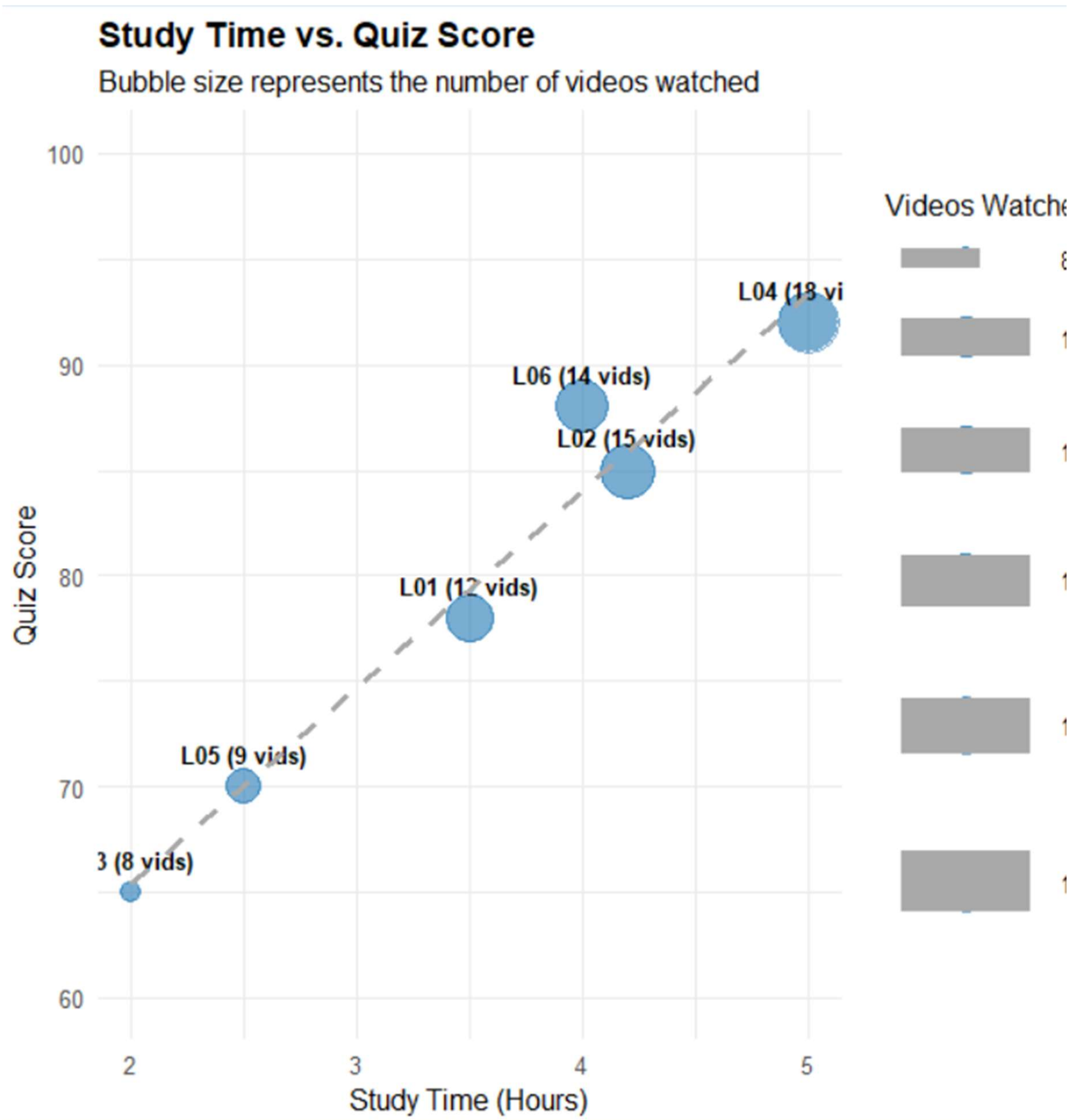
Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.
Code:

```
# Load required libraries

library(ggplot2)

library(dplyr)

library(zoo) # For moving average calculations
```

```
# 1. Import dataset from CSV or build directly
# learning_data <- read.csv("Online_Learning_Activity.csv")

learning_data <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
  Login_Date = c("2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-03-12",
"2025-03-12")
)

# View dataset structure
str(learning_data)

# -----------------------------------------------------------
# 2. Bubble Scatter Plot with Transparency
# -----------------------------------------------------------
ggplot(learning_data, aes(x = Study_Time, y = Quiz_Score, size = Videos_Watched)) +
  geom_point(color = "#1F77B4", alpha = 0.6) +
  geom_text(aes(label = paste0(Student_ID, " (", Videos_Watched, " vids)")),
            vjust = -1.2, size = 3.5, fontface = "bold") +
  geom_smooth(method = "lm", se = FALSE, color = "darkgray", linetype = "dashed") +
  scale_size_continuous(range = c(4, 12), name = "Videos Watched") +
  scale_y_continuous(limits = c(60, 100)) +
```

```
labs(
  title = "Study Time vs. Quiz Score",
  subtitle = "Bubble size represents the number of videos watched",
  x = "Study Time (Hours)",
  y = "Quiz Score"
) +
theme_minimal(base_size = 12) +
theme(plot.title = element_text(face = "bold", size = 14))
```

Output:



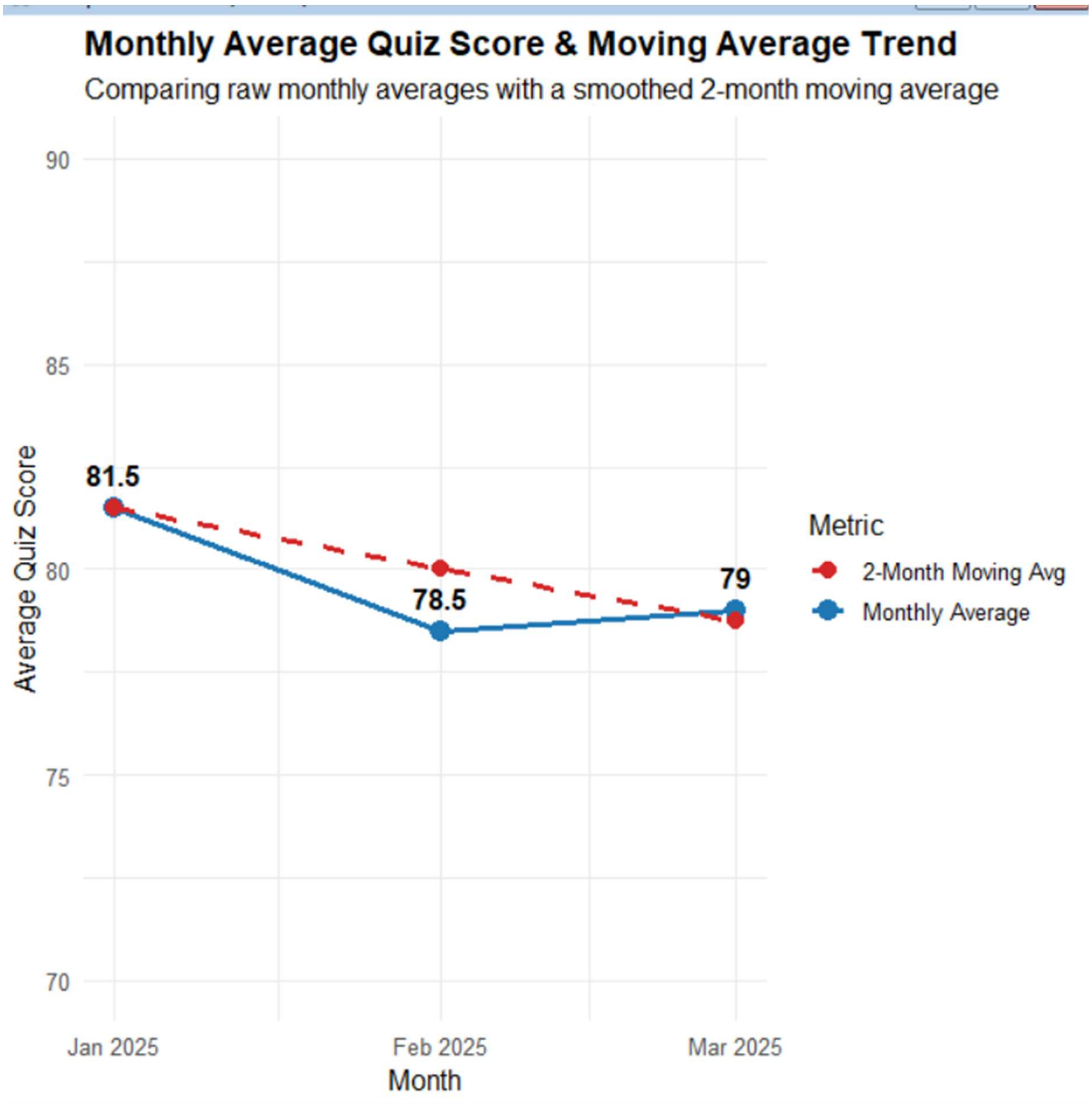
Code:

```
# ===============================================================

# Task 3: Fixed Script (Date Conversion + Monthly Trend + Moving Avg)

# ===============================================================


library(ggplot2)

library(dplyr)


# 1. Dataset

learning_data <- data.frame(

  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),

  Course = c("R", "R", "SQL", "R", "R", "SQL"),

  Quiz_Score = c(78, 85, 65, 92, 70, 88),

  Login_Date = as.Date(c("2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-
03-12", "2025-03-12"))

)


# 2. Calculate Monthly Averages and 2-Month Moving Average safely

monthly_summary <- learning_data %>%

  mutate(Month_Date = as.Date(cut(Login_Date, breaks = "month"))) %>%

  group_by(Month_Date) %>%

  summarise(Avg_Quiz_Score = mean(Quiz_Score), .groups = "drop") %>%

  arrange(Month_Date) %>%

  mutate(

    # Safely compute 2-month trailing moving average manually:

    # Month 1 = Jan score

    # Month 2 = Average of (Jan + Feb)

    # Month 3 = Average of (Feb + Mar)

    Moving_Avg = c(
```

```
    Avg_Quiz_Score[1],
    (Avg_Quiz_Score[1] + Avg_Quiz_Score[2]) / 2,
    (Avg_Quiz_Score[2] + Avg_Quiz_Score[3]) / 2
   )
  )

# View the calculated table
print(monthly_summary)

# 3. Plot Line Chart with Moving Average
ggplot(monthly_summary, aes(x = Month_Date)) +
  # Raw monthly average line
  geom_line(aes(y = Avg_Quiz_Score, color = "Monthly Average"), size = 1.2) +
  geom_point(aes(y = Avg_Quiz_Score, color = "Monthly Average"), size = 4) +
  geom_text(aes(y = Avg_Quiz_Score, label = round(Avg_Quiz_Score, 1)), vjust = -1,
fontface = "bold") +

  # Moving average line
  geom_line(aes(y = Moving_Avg, color = "2-Month Moving Avg"), size = 1.2, linetype =
"dashed") +
  geom_point(aes(y = Moving_Avg, color = "2-Month Moving Avg"), size = 3) +

  scale_x_date(date_labels = "%b %Y", date_breaks = "1 month") +
  scale_y_continuous(limits = c(70, 90)) +
  scale_color_manual(values = c("Monthly Average" = "#1F77B4", "2-Month Moving Avg" =
"#D62728")) +
  labs(
    title = "Monthly Average Quiz Score & Moving Average Trend",
    subtitle = "Comparing raw monthly averages with a smoothed 2-month moving average",
    x = "Month",
```

```
    y = "Average Quiz Score",
    color = "Metric"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Output:



Set 10:

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart

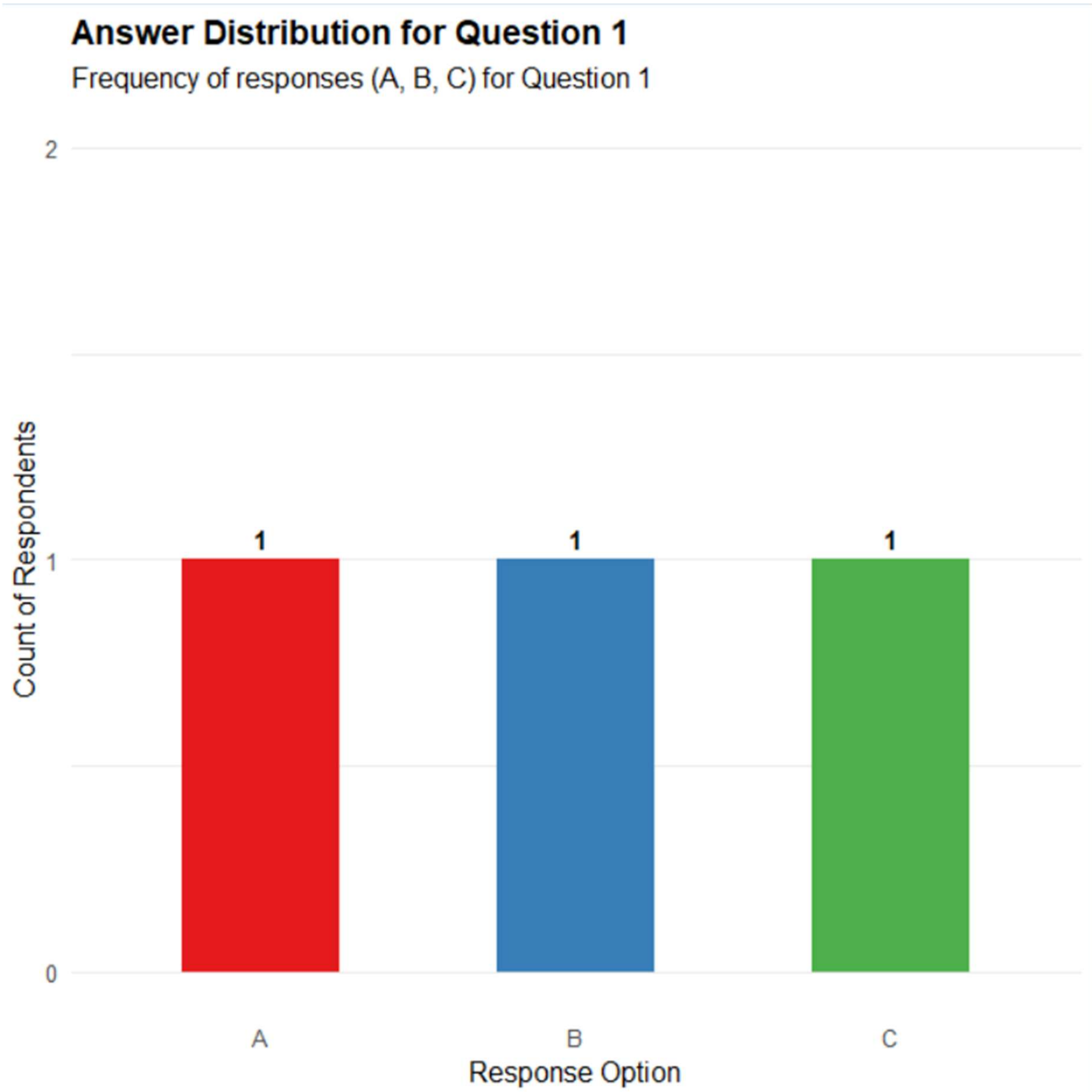
elements.

Code:

```r
# Load required libraries
library(ggplot2)
library(dplyr)
library(tidyr)
library(knitr)

# Create the survey responses dataset
survey_data <- data.frame(
  Survey_ID = factor(c(1, 2, 3)),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

# View dataset
print(survey_data)

# 1. Bar Chart: Answer Distribution for Question 1
q1_summary <- survey_data %>%
  count(Question_1)

ggplot(q1_summary, aes(x = Question_1, y = n, fill = Question_1)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = n), vjust = -0.5, fontface = "bold", size = 4) +
  scale_y_continuous(limits = c(0, 2), breaks = c(0, 1, 2)) +
```

```
scale_fill_brewer(palette = "Set1") +
labs(
  title = "Answer Distribution for Question 1",
  subtitle = "Frequency of responses (A, B, C) for Question 1",
  x = "Response Option",
  y = "Count of Respondents"
) +
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(face = "bold", size = 14),
  panel.grid.major.x = element_blank()
)
```

Output:



2. Generate a stacked bar chart to represent the overall distribution of responses for all three

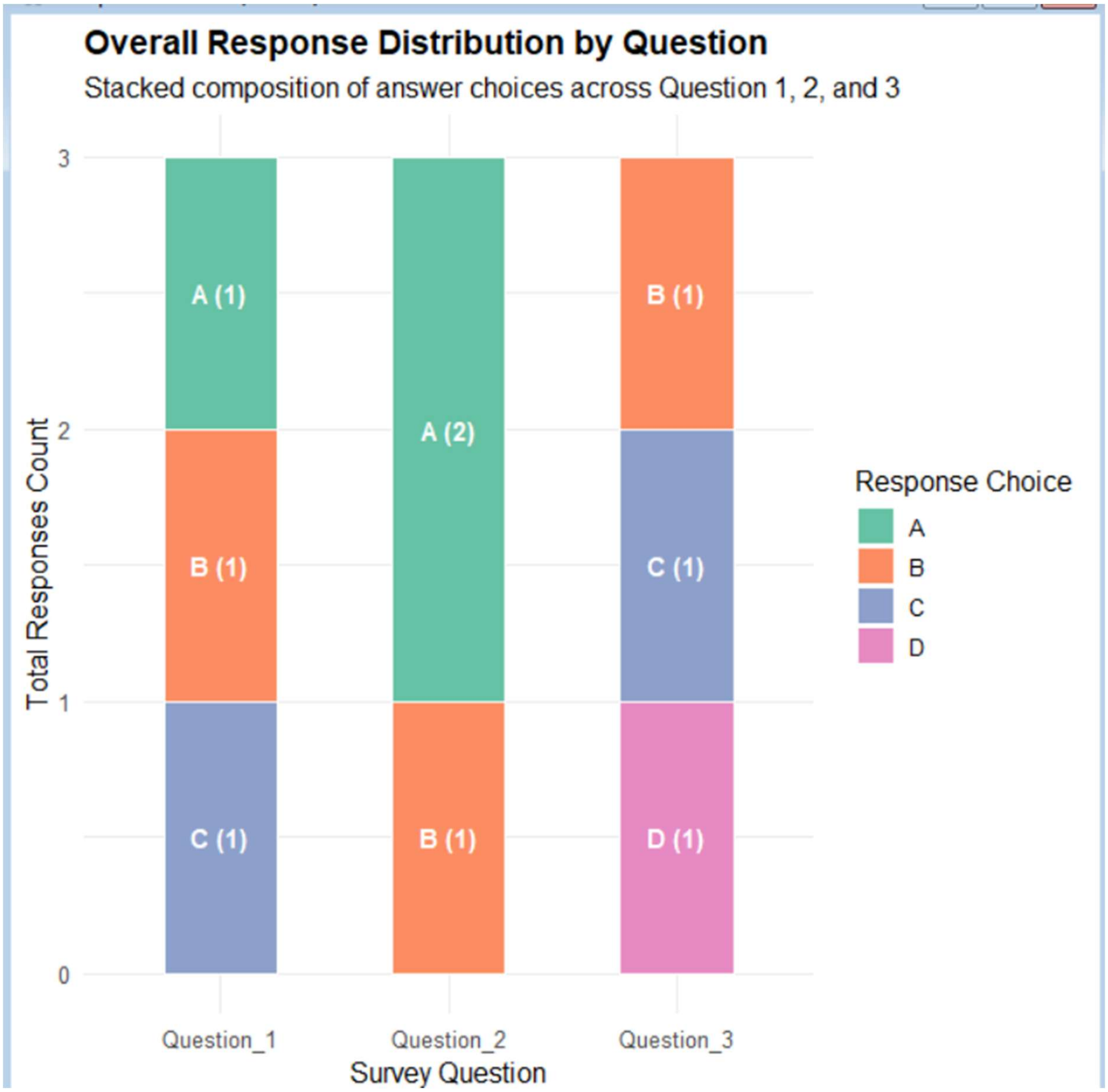
questions.
Code:

```
# Load required libraries

library(ggplot2)

library(dplyr)

library(tidyr)

library(knitr)
```

```
# Create the survey responses dataset
survey_data <- data.frame(
  Survey_ID = factor(c(1, 2, 3)),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

# View dataset
print(survey_data)

# 2. Stacked Bar Chart: Overall Response Distribution across All Questions
survey_long <- survey_data %>%
  pivot_longer(
    cols = c(Question_1, Question_2, Question_3),
    names_to = "Question",
    values_to = "Response"
  ) %>%
  count(Question, Response)

ggplot(survey_long, aes(x = Question, y = n, fill = Response)) +
  geom_col(position = "stack", width = 0.5, color = "white") +
  geom_text(
    aes(label = paste0(Response, " (", n, ")")),
    position = position_stack(vjust = 0.5),
    color = "white",
    fontface = "bold",
    size = 3.8
  ) +
```

```
scale_fill_brewer(palette = "Set2") +
scale_y_continuous(breaks = 0:3) +
labs(
  title = "Overall Response Distribution by Question",
  subtitle = "Stacked composition of answer choices across Question 1, 2, and 3",
  x = "Survey Question",
  y = "Total Responses Count",
  fill = "Response Choice"
) +
theme_minimal(base_size = 12) +
theme(plot.title = element_text(face = "bold", size = 14))
```

Output:



3. Build a table to show the survey response data for each survey. Label the table elements.

Code:

```
# Load required libraries

library(ggplot2)

library(dplyr)

library(tidyr)

library(knitr)


# Create the survey responses dataset
```

```r
survey_data <- data.frame(
  Survey_ID = factor(c(1, 2, 3)),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

# View dataset
print(survey_data)

# 3. Table Output
survey_table <- survey_data %>%
  rename(
    `Survey ID` = Survey_ID,
    `Question 1` = Question_1,
    `Question 2` = Question_2,
    `Question 3` = Question_3
  )

# Display clean table
kable(survey_table, caption = "Survey Response Summary Table", align = "c")
```

Output:

```
Table: Survey Response Summary Table

| Survey ID | Question 1 | Question 2 | Question 3 |
|:---------:|:----------:|:----------:|:----------:|
|     1     |     A      |     B      |     C      |
|     2     |     B      |     A      |     D      |
|     3     |     C      |     A      |     B      |
>
```